



OPEN Resistome analysis of wastewater treatment plants in Agadir city, Morocco, using a metagenomics approach

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Water scarcity has evolved into a pressing global issue, significantly impacting numerous regions worldwide. The use of treated wastewater stands out as a promising solution to this problem. However, the proliferation of various contaminants, primarily Antimicrobial Resistance Genes (ARGs), poses a significant challenge to its safe and sustainable use. In this study, we assessed the composition and abundance of 373 ARGs, corresponding to 31 different classes of antibiotics, in six wastewater treatment plants (WWTP) in Agadir city of Morocco. Influent and effluent samples were collected during the months of February and July in 2020, in addition to samples from the Atlantic Ocean. In total, 223 ARGs were uncovered, highlighting in particular resistance to aminoglycoside, macrolide lincosamide, beta-lactamase, chloramphenicol, sulfonamide, tetracycline, and other antibiotics. The mechanisms of action of these ARGs were mainly antibiotic inactivation, antibiotic target alteration, efflux pump and cellular protection. Mobile genetic elements (MGEs) were detected at high levels their co-occurrence with ARGs highlights their involvement in the acquisition and transmission of ARGs in microbial communities through horizontal gene transfer. While many wastewater treatment methods effectively reduce a large proportion of gene material and pathogens, a substantial fraction of ARGs and other contaminants persist in treated wastewater. This persistence poses potential risks to both human health and the environment, warranting the need of more effective treatment strategies.

Keywords Water scarcity, Wastewater, Antimicrobial resistance genes, Microbiome, Public Health

Water scarcity has become an increasingly urgent global issue, deeply impacting regions around the world and affecting billions of lives¹. This shortage stems from a complex interplay of factors, such as population expansion, climate change, ineffective water management, and pollution. Climate change is intensifying the issue of water scarcity by disrupting rainfall patterns and amplifying the frequency and severity of droughts and floods². These changes disrupt the natural water cycle, leading to irregularities in water availability and distribution³.

In addressing water scarcity, various strategies are employed worldwide. Among these strategies, one particularly crucial approach is the reuse of treated wastewater. This method involves treating wastewater to remove contaminants, transforming it into a valuable resource for non-potable purposes. Advanced treatment technologies ensure that recycled water meets quality standards, making it safe for applications such as irrigation, industrial processes, and even toilet flushing⁴. By employing recycled water for these activities, pressure on traditional water sources is reduced, contributing to water conservation efforts. Embracing wastewater reuse as a sustainable solution supports the efficient management of water resources, promoting environmental and economic sustainability⁵.

However, the reuse of treated wastewater for various purposes, while beneficial in addressing water scarcity, raises concerns about the emergence of new contaminants, particularly antimicrobial resistance (AMR) genes. It can contain residues of pharmaceuticals, including antibiotics, and other antimicrobial agents. These antimicrobial substances can exert selective pressure on bacteria present in the environment, promoting the

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development and spread of resistance, which can be transferred to other bacteria through horizontal gene transfer mechanisms⁶. The presence of AMR genes in recycled wastewater poses risks to public health, as it can lead to the dissemination of antibiotic-resistant bacteria and compromise the effectiveness of antibiotics in medical treatment. A recent study examined the metagenomic insights into the wastewater resistome in large-scale wastewater treatment plants (WWTPs) across Moscow city revealed that treated effluent continued to contain AMR genes, with a notable prevalence of those associated with beta-lactamases. Among these ARGs, *ampC* was identified as the most abundant, even after the purification process⁷. Another study indicates that wastewater treatment effectively eliminates the hosts of antibiotic resistance genes, leading to a subsequent reduction in the presence of harbored antibiotic resistance genes⁸.

To address the emergence of those new contaminants, comprehensive monitoring and treatment protocols are essential. This involves implementing advanced technologies capable of removing a wide range of contaminants. Additionally, ongoing surveillance programs can track the presence and prevalence of AMR genes in wastewater, assessing potential risks to public health and the environment^{9,10}.

In this context, Morocco faces significant challenges regarding water scarcity, particularly in regions with limited freshwater resources and growing water demand. The country's arid and semi-arid climate exacerbates the problem, leading to water stress and shortages, especially during prolonged droughts. To address water scarcity, Morocco has been increasingly exploring the reuse of treated wastewater as sustainable solutions. However, the lack of comprehensive studies evaluating the quality of treated wastewater poses challenges, especially concerning the potential risks associated with antimicrobial resistance. Comprehensive studies evaluating the presence and persistence of AMR genes in treated wastewater are crucial for informing decision-making and implementing effective mitigation strategies. This can enhance water security, protect public health, and contribute to global efforts to combat AMR. This study presents a spatiotemporal characterization of the resistome and microbiome across different wastewater treatment plants in Agadir city during two distinct seasons, with samples collected from the ocean. Additionally, it includes a comparative evaluation of the efficacy of various purification treatments deployed at each facility.

Methods

Samples collection

In February and July 2020, samples of both untreated wastewater and treated wastewater were collected from six wastewater treatment plants. These plants include M'ZAR, ANZA, AOURIR, DRARGA, TIZNIT, and AIT BAHA WWTPs (Fig. 1). A volume of 1 L was collected from both untreated and treated water samples.

The wastewater treatment plants employ various processes for wastewater treatment, as detailed in Table 1. The anaerobic decantation process relies on bacteria to break down organic matter in the absence of oxygen, producing biogas, and is suited for high-strength wastewater. Activated sludge process uses aerobic microorganisms to degrade organic material in the presence of oxygen, typically in municipal wastewater treatment. Lamellar decantation enhances sedimentation efficiency using inclined plates, allowing suspended solids to settle more quickly. Infiltration-percolation on sand involves filtering wastewater through sand beds, offering a natural filtration method. Naturally aerobic lagoons are shallow ponds where microorganisms, aided by sunlight and wind, treat wastewater through aerobic processes, but they require large land areas and extended retention times.

Sampling was exclusively conducted in February for the AIT BAHA station, as the facility underwent temporary closure in July for maintenance purposes. Additionally, water samples were procured from two beaches in July period, ANZA and SIDI TOUAL. To serve as an external control for the experiment, a sample from the TAMZERGOUT river was collected. Subsequently, all 26 gathered samples were carefully transported to the laboratory, ensuring preservation in a cooler set at 4 °C.

Physico-chemical parameters

In-situ measurements of physicochemical parameters encompassed temperature, pH, conductivity, salinity, and dissolved oxygen. These measurements were conducted utilizing a multiparameter device, the Multi 340i by Geotech Environmental Equipment. Laboratory-scale measurements involved turbidity and chemical oxygen demand (COD). Turbidity was assessed using a turbidimeter (LOVIBOND TB300 IR), while the Hanna instrument micro method (Hanna Instruments, USA) was employed for COD determination. The quantification of Calcium, Chlorides, Magnesium, Water Hardness, and content of Carbonates and Bicarbonates (TAC) employed volumetric titration methods¹¹. Calcium and Magnesium, which contribute to water hardness, were measured together using EDTA titration, where EDTA binds with these ions to form a stable complex. Chlorides were quantified by argentometric titration, involving the reaction of chloride ions with silver nitrate. The content of Carbonates and Bicarbonates was determined by acid-base titration, where the sample was titrated with a strong acid to a specific pH, indicating the total alkalinity. Heavy metal analysis took place at the Mitra S. K Morocco SARL laboratory (Mitra S.K, Agadir, Morocco). The array of heavy metals measured encompasses Zinc, Antimony, Cadmium, Aluminum, Lead, Copper, Cadmium, Iron, Manganese, Strontium, Arsenic, and Mercury. To measure heavy metals such as Zinc, Antimony, Cadmium, Aluminum, Lead, Copper, Iron, Manganese, Strontium, Arsenic, and Mercury, Atomic Absorption Spectroscopy (AAS) is used. First, the sample is prepared by digesting it with strong acids to release the metals into solution. The sample is then analyzed using AAS, where it is atomized and exposed to light. Each metal absorbs light at specific wavelengths, and the amount of light absorbed is proportional to the metal's concentration in the sample. Calibration curves, created from standards of known concentrations, are used to quantify the metal concentrations accurately¹².

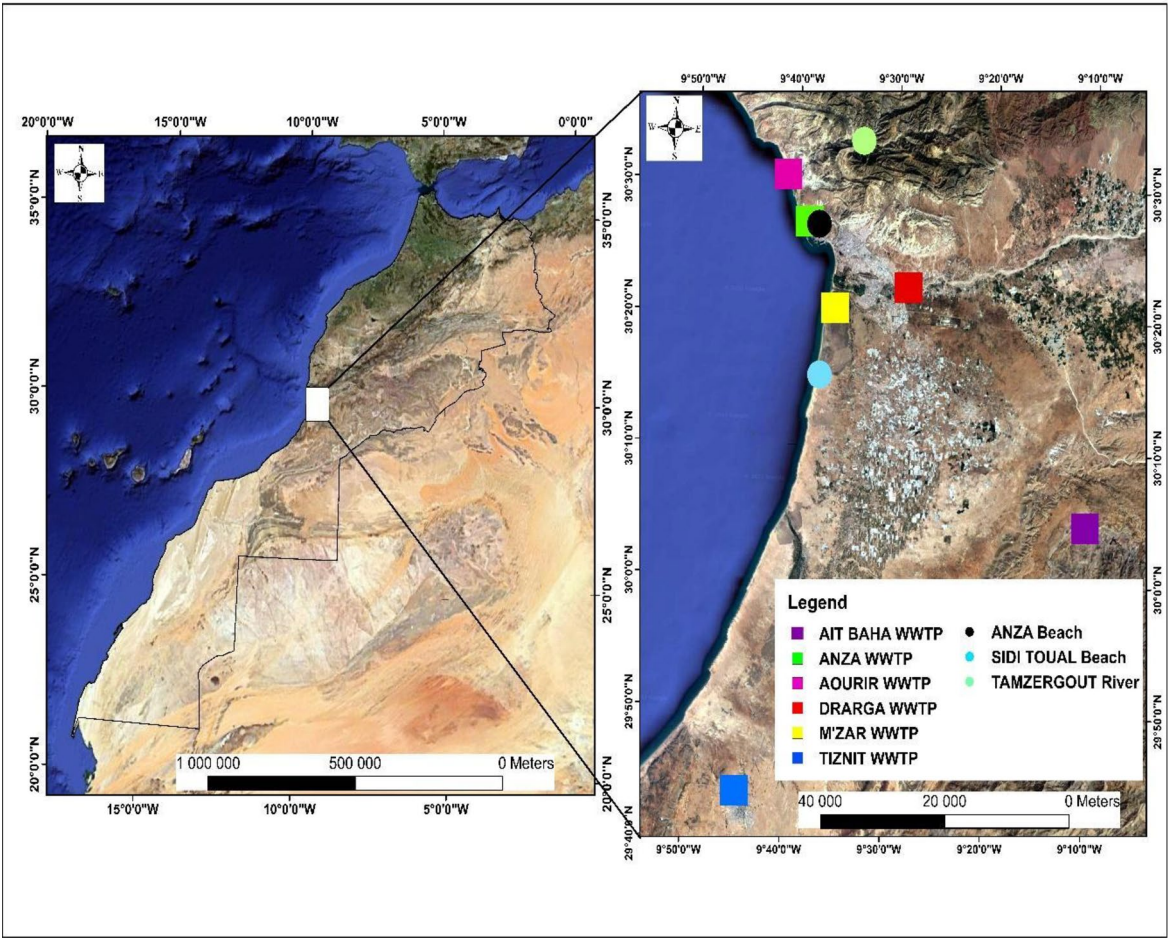


Fig. 1. Map displaying sampling sites created using ArcGIS (version 3.3; <https://www.esri.com/en-us/arcgis/products/arcgis-pro/overview>).

WWTP	Type of treatment	Role of treatment
M'ZAR (MZ)	Anaerobic decantation (A.decantation)	Eliminates a significant portion of suspended matter.
AOURIR (AR)	Activated sludge process (A.sludge)	Wastewater is mixed and agitated with highly active bacteriological sludge. Degradation takes place under aerobic conditions, utilizing a combination of purifying microorganisms and the influent to be treated
ANZA (AZ)	Lamellar decantation (L.decantation)	Clears water by isolating and removing suspended particles
DRARGA (DR)	Infiltration-Percolation on sand (In-Per on sand)	This involves channeling previously settled wastewater through a porous mass of materials, serving as a substrate for purifying microorganisms.
TIZNIT (TZ) and AIT BAHA (AB)	Naturally Aerobic Lagoons (N.Aer Lagoons)	Wastewater resides within basins for a defined period, during which bacterial action within the water degrades pollutants. Solar radiation contributes to the destruction of certain microorganisms

Table 1. Treatments used in studied WWTPs.

DNA extraction, Metagenomic Sequencing and Bioinformatic Analysis

Initially, the wastewater underwent a pre-treatment. Duplicate volumes of 90 ml were subjected to centrifugation at 4000 g, held at 4 °C for 20 min. Subsequently, the resulting pellet was recovered, subjected to a wash with phosphate-buffered saline (PBS), and processed using the QIAamp PowerFecal ProDNA Kit (Qiagen, Hilden, Germany) in accordance with the kit's extraction protocol. For the water samples collected from ANZA and SIDI TOUAL beaches, they underwent a filtration process of 1 L utilizing a 0.45 µm filter. Following filtration, the filter was segmented into smaller sections and then subjected to treatment using the mentioned extraction kit. Quantification of the extracted DNA was achieved through both a DeNovix DS-11 spectrophotometer and Qubit (HS ds DNA test kit).

To check the quality of the extracted DNA and verify that it does not contain PCR inhibitors, 16S PCR was performed. The 16S rRNA gene was amplified using universal primers *Fd1* (5'-AGAGTTTGATCCTGGCTCA G-3') and *rP2* (5'-ACGGCTACCTTGTTCAGACTT-3')¹³. PCR amplification was carried out in a final volume of 15 µl containing 7,8 µl nuclease Free water, 3 µl of MyTaq Reaction Buffer, 0,2 µl MyTaq™ DNA Polymerase (MyTaq™ DNA Polymerase, Bioline Reagents Ltd, Meridian Bioscience) and 1 µl of primers (10 pmoles) and 3 µl (10ng) of DNA template. The PCR started with an initial denaturation of 2 min at 95 °C, followed by 30 denaturation cycles at 95 °C for 15 s, hybridization at 52 °C for 30 s, extension at 72 °C for 1 min, and a final elongation step for 7 min at 72 °C. PCR products (10 µL of PCR products) were transferred to a 1% agarose gel in TBE migration buffer (Tris, Borate, EDTA). The Ion AmpliSeq™ Pan-Bacterial Research Panel from Thermo Fisher Scientific was employed to characterize the resistome and microbiome of samples. The panel contained two pools. The first pool contained 269 amplicons to track 21 specific species and 716 amplicons to detect 364 antibiotic resistance genes belonging to 31 different antibiotic classes. The second pool contained 24 amplicons that were used to characterize the 16 S profile. The library was prepared with ION CHEF (Ion Chef™ Instrument - Thermo Fisher Scientific) using the Ion Ampliseq kit for chef DL8 (Ion AmpliSeq™ Kit for Chef DL8 - Thermo Fisher Scientific). The concentration was set at 10 ng in a 15 µL reaction volume. For automated library preparation, R&D recommends cycling the two pools separately under the following conditions:

- Pool 1 (SpID/AMR): 985 amplicons with 15 cycles and a 4-minute annealing time.
- Pool 2 (16 S): 24 amplicons with 21 cycles and a 4-minute annealing time.

For DL8 adjustments, an additional cycle is typically added to the target amplification, but the pools are run on separate automated Chef DL8 runs.

The library was subsequently quantified by the Ion Library TaqMan Quantitation Kit (Ion Library TaqMan™ Quantitation Kit - Thermo Fisher Scientific). The samples were prepared and subsequently loaded onto a 540-chip using an Ion Chef system. After loading, the chips were sequenced using the Ion GeneStudio S5 system (Ion GeneStudio™ S5 System - Thermo Fisher Scientific). Bioinformatics analysis was performed using the Torrent Suite plugin under the name « Pan Bacterial Analysis ». This plugin is available on the ThermoFisher Scientific Plugin store.

qPCR of additional antimicrobial resistance genes

The analysis panel employed did not cover certain crucial genes that present a notable threat to public health. The focus was on genes such as *blaNDM*, *blaCTX*, *blaVIM*, *blaOXA-48*, *blaKPC*, *blaSHV*, *blaIMP*, and *mcr-1*. To standardize the gene copy counts, 16 S qPCR was conducted¹⁴. The primers used to detect these genes are listed in the Table 2.

qPCR was performed on an AriaMx Agilent (AriaMx Real-time PCR System, Agilent) using Syber Green, and according to MIQE guidelines¹⁵. Each qPCR run was performed in triplicate in a final volume of 10 µl which contained 5 µl of Power SYBR™ Green PCR Master mix (Applied Biosystems, Carlsbad, CA), 2 µl of primers (5pmol/µl) and 3 µL (2ng for 16 S gene and 5ng for the ARG genes) of DNA. Reactions were conducted using the Agilent Aria 1.8 system with the following thermal profile: AmpliTaqGold™ Polymerase activation at 95 °C for 10 min, followed by 40 cycles of denaturation at 95 °C for 15 s. The annealing/extension temperatures were set as follows: *blaNDM* at 58 °C, *blaOXA-48* at 55 °C, *blaKPC* at 58 °C, *blaIMP* at 56 °C, *blaVIM* at 58 °C, *blaCTX* at 57 °C, *mcr-1* at 58 °C, 16 S at 60 °C, and *blaSHV* at 64 °C for 1 min. A standard curve was established through a 10-fold serial dilution of control bacteria hosting the target genes. To determine the specificity of the amplification, a melting curve was recorded by increasing the temperature from 60 °C to 95 °C. Data analysis was performed using Agilent Aria 1.8 software.

Statistical analysis

Principal Component Analysis (PCA) of physicochemical parameters was conducted utilizing the FactoMiner package in the RStudio software (version 4.3.3). Figures were generated employing the ggplot2, tidy, and tidyverse packages. Alpha diversity (index = Shannon), Beta diversity analysis (method = Bray), Permutational Multivariate Analysis of Variance (PERMANOVA), and Analysis of Similarities (ANOSIM) tests were executed through the vegan package within RStudio¹⁶. The abundance data was first transformed using a log transformation (log1p), and then it was subjected to min-max scaling to normalize the values between 0 and 1. The outcomes of these analyses were then visually represented using the ggplot2 package.

Canonical Correspondence Analysis (CCA) was performed using the vegan package in R. This analysis enabled the exploration of relationships between physicochemical parameters and antibiotic resistance gene (ARG) composition in the dataset. Spearman correlation analysis among Mobile Genetic Elements (MGE), Antibiotic Resistance Genes (ARG), microbiome and species was executed utilizing the ggplot2 and dplyr libraries in R. The resistome and microbiome co-occurrence network analysis was executed utilizing the igraph, readr, ggraph, dplyr, and ggforce libraries. This analytical approach involved establishing a correlation threshold of 0.6, enabling the identification of strong associations among the data points.

Results

Physicochemical parameters

Sampling of wastewater was conducted in February (Feb) and July (Jul) 2020 with the purpose of evaluating seasonal fluctuations in physicochemical parameters at designated stations (Table 3). The choice of February and July is dictated by the fact that the region is characterized by a semi-arid to sub-desert climate; most of the rainfall is limited to the period between November and March, while the dry season lasts from May to October¹⁷. Throughout both sampling periods, effluent temperatures were maintained below 30 °C. In February,

Target Gene	Primer	Sequence (5'-3')	Reference
<i>blaNDM</i>	<i>blaNDM_SW-F</i>	TTGGCCTTGCTGTCCTTG	1
	<i>blaNDM_SW-R</i>	ACACCAGTGACAATATCACCG	
<i>blaOXA-48-like</i>	<i>blaOXA-48-like_SW-F</i>	TATCGGCTGTGTTTTTGGTG	2
	<i>blaOXA-48-like_SW-R</i>	CAACTTTTGTCTTCTGCCATTC	
<i>blaKPC</i>	<i>blaKPC_SW-F</i>	ACTGTAAGTTACCGCGCTGA	2
	<i>blaKPC_SW-R</i>	AAGAAAGCCCTTGAATGAGCT	
<i>blaIMP</i>	<i>blaIMP_SW-F</i>	GGAATAGAGTGGCTTAATTCTCRATC	3
	<i>blaIMP_SW-R</i>	AGGYARCCAAACCACTAWGTTATC	
<i>blaVIM</i>	<i>blaVIM_SW-F</i>	CTCGCGGAGATTGAAAAGCA	2
	<i>blaVIM_SW-R</i>	CGRTCGTCATGRAAGTGCG	
<i>blaCTX-M</i>	<i>blaCTX-M_SW-F</i>	TGATACCACTTCACCTCGGG	2
	<i>blaCTX-M_SW-R</i>	GCTTTACCCAGCGTCAGATT	
<i>bla-SHV</i>	<i>bla-SHV_SW-F</i>	CGCTTTCCCATGATGAGCACCTTT	4
	<i>bla-SHV_SW-R</i>	TCCTGCTGGCGATAGTGGATCTTT	
<i>mcr-1</i>	<i>mcr-1_SW-F</i>	GGGCCTGCGTATTTTAAGCG	6
	<i>mcr-1_SW-R</i>	CATAGGCATTGCTGTGCGTC	
<i>ermB</i>	<i>ermB_SW-F</i>	TGAATCGAGACTTGAGTGTGCAA	5
	<i>ermB_SW-R</i>	GGATTCTACAAGCGTACCTT	
16 S	16S_SW-F	CCTACGGGAGGCAGCAG	7
	16S_SW-R	ATTACCGCGGCTGCTGG	

Table 2. Primers used for quantitative PCR of other resistance genes. 1. Monteiro, J., Widen, R. H., Pignatari, A. C. C., Kubasek, C. & Silbert, S. Rapid detection of carbapenemase genes by multiplex real-time PCR. *Journal of Antimicrobial Chemotherapy* **67**, 906–909 (2012). 2. Guzman-Otazo, J. et al. Diarrheal bacterial pathogens and multi-resistant enterobacteria in the Choqueyapu River in La Paz, Bolivia. *PLoS One* **14**, 1–20 (2019). 3. Flach, C. F., Hutinel, M., Razavi, M., Åhrén, C. & Larsson, D. G. J. Monitoring of hospital sewage shows both promise and limitations as an early-warning system for carbapenemase-producing Enterobacterales in a low-prevalence setting. *Water Res* **200**, (2021). 4. Xi, C. et al. Prevalence of antibiotic resistance in drinking water treatment and distribution systems. *Appl Environ Microbiol* **75**, 5714–5718 (2009). 5. Alexander, J., Bollmann, A., Seitz, W. & Schwartz, T. Microbiological characterization of aquatic microbiomes targeting taxonomical marker genes and antibiotic resistance genes of opportunistic bacteria. *Sci Total Environ* **512–513**, 316–325 (2015). 6. Hembach, N. et al. Occurrence of the *mcr-1* colistin resistance gene and other clinically relevant antibiotic resistance genes in microbial populations at different municipal wastewater treatment plants in Germany. *Front Microbiol* **8**, (2017). 7. Muyzer, G., de Waal, E. C. & Uitterlinden, A. G. Rapid Lipid Biomarker Analysis for Quantitative Assessment of Microbial Community Composition and Activity. *Applied and Environmental Microbiology* **59**, 695–700 (1993).

temperatures remained consistently within the range of 18–20 °C, showing negligible disparities between influent (IF) and effluent (EF) values. Conversely, in July, a slight rise in temperature was observed, ranging from 19 to 26 °C. pH levels fluctuated between 7 and 8 during February and 6–8 in July. Notably, all recorded values remained below 8.5, indicating compliance with acceptable thresholds for direct discharge into the receiving environment.

Significant differences in conductivity were observed between the two sampling periods, with certain stations surpassing specified limits. Of particular concern was the ANZA station, which recorded notably high values exceeding the prescribed thresholds. This could potentially be explained by the fact that the ANZA station receives salt-laden waters originating from fish processing industries located in its area. Dissolved oxygen levels were higher in July, accompanied by elevated turbidity in influents. COD indicated a reduction post-treatment, although certain effluents slightly exceeded discharge limits. While influent COD levels were notably high, they decreased following treatment. However, select effluents marginally surpassed the direct discharge threshold of 500 mg/L. Calcium, chlorides, magnesium, water hardness, and carbonate and bicarbonate concentrations were relatively moderate to low based on the sampled data from both periods. Analysis of heavy metals revealed trace amounts of antimony not exceeding 0.2 mg/L. The origin of antimony is primarily due to anthropogenic inputs, as it is largely absent from the geological strata. Antibiotic resistance genes occur naturally in various environments in low abundance. However, studies have shown that their abundance increase in the presence of pollutants such as heavy metals, crude oil and sewage^{18,19}.

Principal Component Analysis revealed correlations among various parameters, indicating correlation between temperature, chlorides, salinity, and calcium, alongside TAC, COD, turbidity, and conductivity. Additionally, pH demonstrated positive correlations with dissolved oxygen, total hardness, and magnesium content. These correlations accounted for 34.3% and 22.6% of the total variance, respectively (Fig. 2A). Furthermore, analyzing temporal shifts between February and July showed significant disparities due to seasonal

Sample	pH	Temperature (°C)	Dissolved O ₂ (mg/L)	Conductivity (µS/cm)	Salinity (psu)	Turbidity (NTU)	COD (mg/L)	Calcium (mg/L)	Total Hardness (°F)	Magnesium (mg/L)	TAC (mg/L)	Chlorides (mg/L)	Antimony (Sb) mg/L
IF_MZ_Feb20	7.44±0.015	19.35±0.35	0.25±0.025	3630±0	1.8±0	266.5±5.8	2770±50	32.06±0.5	32±0	58.33±3.44	265.41±0	421.15±2.8	0.14±0
EF_MZ_Feb20	7.32±0.02	19.6±0.2	0.78±0.02	4330±10	2.2±0	394.2±0.5	435±95	160.31±0	60±0	48.61±2.5	284.76±0	592.72±4.8	0.11±0.01
IF_AR_Feb20	7.59±0.01	20.4±0.3	0.32±0.025	2455±15	1.1±0	329.35±3.25	2660±420	48.09±1.3	24±0	29.17±0.56	201.82±0	265.17±2.23	0±0
EF_AR_Feb20	7.5±0.015	19.15±0.05	3.52±0.015	1800.5±0.5	0.7±0	3.4±0.68	240±150	128.25±2.5	40±0	19.44±0.89	69.12±0.3	233.97±1.8	0.2±0.001
IF_AZ_Feb20	7.55±0.01	21.6±0.1	0.69±0.04	4825±5	2.5±0	302.3±5.7	2580±250	96.19±0	40±0	38.89±0.2	236.38±2.4	330.68±4.3	0.17±0.002
EF_AZ_Feb20	7.55±0.015	20.85±0.05	0.25±0.05	2004±6	0.85±0.05	213.7±1	535±105	80.16±1.2	40±0	48.61±0.56	175.56±0	137.26±0	0.12±0
IF_DR_Feb20	7.62±0.005	18.9±0.1	0.35±0.15	2545±5	1.2±0	359.4±20.6	2190±30	80.16±1.5	32±0	29.17±0.45	239.15±0	202.77±2.5	0.19±0
EF_DR_Feb20	7.84±0.01	18.85±0.05	1.31±0.1	3920±30	2±0	47.95±1.325	375±125	64.12±0	40±0	58.33±1.2	226.7±4.9	233.97±2.5	0.11±0.01
EF_AB_Feb20	7.69±0.03	18.55±0.05	0.24±0.04	3340±10	1.6±0	520.4±16.6	2570±10	80.16±1.3	1600±0	3840.19±23	252.97±2.3	140.38±0	0.13±0.001
IF_AB_Feb20	8.01±0.005	18.7±0.1	1.11±0.1	2750±10	1.3±0	142.25±2.75	780±120	32.06±0.4	1600±0	3869.36±21.5	239.15±2.4	109.19±0	0.18±0
IF_TZ_Feb20	7.39±0.01	18.25±0.05	0.4±0.01	3205±5	1.6±0	383.2±5	2565±355	64.12±1.2	40±0	58.33±2.38	309.64±0	296.36±3.3	0±0
EF_TZ_Feb20	7.58±0	18.6±0	0.74±0.215	2540±10	1.2±0	94.38±0.515	245±45	32.06±0.5	40±0	77.78±4.5	182.47±0	262.05±3.6	0.12±0.002
IF_MZ_Jul20	7.42±0.055	26.5±0.4	0.2±0.045	4200±30	2.2±0	467.9±19.7	955±135	128.25±0	60±0	68.05±3.4	297.2±2.5	686.31±2.4	0±0
EF_MZ_Jul20	7.61±0.05	26.55±0.05	0.58±0.02	4005±55	2±0.1	683.5±0.9	525±5	80.16±0.18	60±0	97.22±0	269.56±3.44	623.92±5.5	0.12±0.001
IF_AR_Jul20	7.18±0.035	25.6±1.2	0.41±0.155	1878.5±6.5	0.8±0	657.9±3.3	975±125	112.22±3.65	40±0	29.17±0.7	138.23±0.34	171.58±1.48	0.13±0.002
EF_AR_Jul20	7.37±0.035	25.4±1.3	3.34±0.275	1911.5±4.5	0.8±0	16.19±0.235	850±40	128.25±2.34	40±0	19.44±0.32	80.18±1.23	218.33±2.75	0±0
IF_AZ_Jul20	6.73±0.075	25.45±1.35	0.52±0.255	16655±105	9.85±0.05	1002.5±2.5	1015±15	48.09±3.6	120±0	262.49±2.55	291.67±4.23	1310.23±2.5	0.2±0.01
EF_AZ_Jul20	6.65±0.035	25.5±1.1	0.78±0.345	15960±20	9.4±0	1065.2±5	5300±200	144.28±5.2	160±0	301.38±3.38	311.03±0	1435.02±0	0.21±0.02
IF_DR_Jul20	7.55±0.02	19.25±0.05	1.71±0.045	2910±10	1.4±0	697.2±11.1	1550±100	80.16±0.11	32±0	29.17±0.7	302.73±0	187.18±1.7	0.12±0
EF_DR_Jul20	7.07±0.005	19.45±0.05	3.82±0.085	2765±5	1.3±0	37.4±0.62	510±280	160.31±2.4	68±0	68.05±3.1	161.73±2.55	202.77±0	0.14±0.01
IF_TZ_Jul20	7.52±0.02	18.9±0.1	1.47±0.1	2785±15	1.3±0	498.55±4.55	1115±155	80.16±0.12	40±0	48.61±2.1	345.59±4.3	249.57±1.4	0.15±0.02
EF_TZ_Jul20	7.93±0.015	19.05±0.05	3.63±0.05	2645±5	1.2±0	79.43±1.03	302.5±97.5	80.16±0	48±0	68.05±2.2	236.38±2.9	265.17±2.5	0.12±0.01
ANZA BEACH	8.08±0.025	23.65±0.25	3.38±0.045	5525±5	36.7±0	0±0	1425±25	320.62±3.44	460±0	923.59±0	45.62±0	11448.9±15	0±0
SIDI TOUAL BEACH	8.07±0.005	23.7±0.1	3.43±0.04	5510±10	36.6±0	0.92±0.57	1370±200	240.47±3.2	420±0	874.98±0	41.47±0	11417.7±13	0±0

Table 3. Physicochemical parameters of samples.

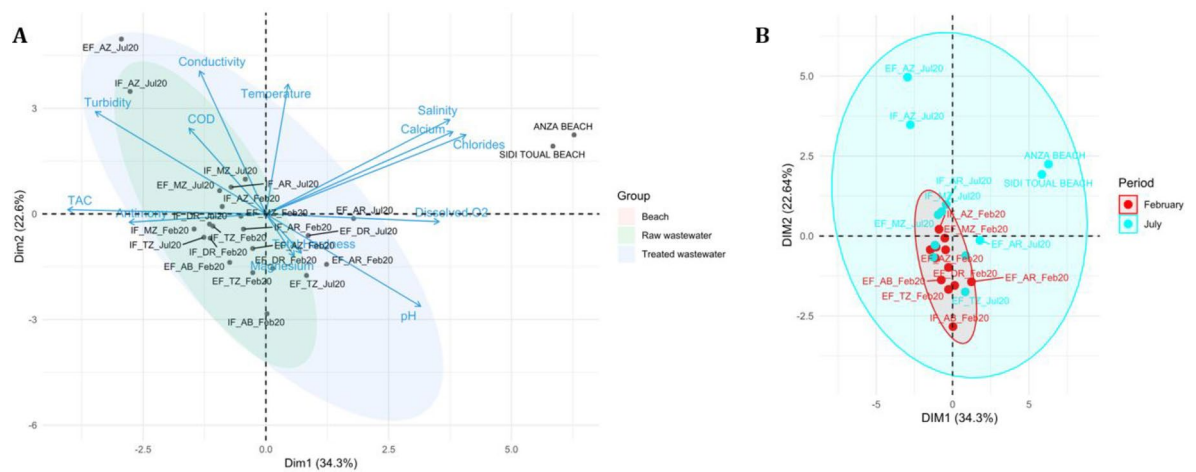


Fig. 2. Principal Component Analysis of physicochemical parameters.

variables such as inputs of rain water experienced only in February and/or touristic activities that are more common in July (Fig. 2B).

Resistome analysis

A sum of 105,853,819 reads were recorded, comprising 1,373,434 reads related to Antibiotic Resistance Genes and 88,642,264 reads that were successfully aligned with the 16 S reference. A cumulative count of 31 distinct antibiotic drug classes and 223 antibiotic resistance genes were documented. The relative abundance of each drug class and ARG is expressed as copies per 1 million 16 S equivalent cells¹⁴. Among the 31 drug classes, the most abundant three drug classes are Aminoglycosides, Lincosamide, Macrolide and Streptogramins (MLS), in addition to Macrolide antibiotics (Fig. 3). The corresponding ARGs linked with these drug classes consistently appear in all samples. The relative abundances of these ARGs encompass a range from 535 to 726,222 copies per 1 million 16 S equivalent cells.

In the February timeframe, when analyzing station data, the input of TIZNIT station stands out with a notable count of 726,222 copies of aminoglycoside antibiotics. Following closely is the ANZA input station, which records 559,743 copies of macrolide antibiotics. Meanwhile, the input of M'ZAR station and AOURIR station exhibit relatively lower abundance in drug classes, with 47,504 copies of MLS and 61,379 copies of macrolide antibiotics, respectively. During the same period, in terms of plant effluents, the ANZA and DRARGA plants displayed the most significant levels of abundance. The ANZA plant exhibited 305,153 copies of macrolide antibiotics, while the DRARGA plant showcased 283,028 copies of aminoglycoside antibiotics. In contrast, the M'ZAR and AOURIR stations demonstrated a comparatively lower relative abundance, with 10,226 copies and 87,475 copies of aminoglycoside antibiotics, respectively. In the month of July, the DRARGA and M'ZAR inlets registered the most notable levels of abundance, with 386,153 and 314,768 copies of aminoglycoside antibiotics, respectively. Conversely, the ANZA and TIZNIT locations documented the lowest figures, displaying 70,102 and 172,359 copies of aminoglycoside antibiotics, respectively. In terms of station outputs, significant figures were noted at the M'ZAR and TIZNIT stations, with 301,468 copies of aminoglycoside antibiotics and 134,790 copies of lincosamide antibiotics recorded, respectively.

In contrast, the AOURIR and ANZA stations recorded significantly lower levels of aminoglycoside antibiotics, with 35,328 copies at AOURIR and 85,754 copies at ANZA. A related study by Lira et al. using metagenomic analysis of an urban resistome before and after wastewater treatment, detected a substantial presence of macrolide resistance genes²⁰. Similarly, another study on the community composition and resistome in a cold climate wastewater treatment plant identified high levels of genes linked to tetracycline resistance, peptide antibiotics, and macrolides²¹. Additionally, Rodriguez et al.'s analysis of urban wastewater resistome and mobilome highlighted the prevalence of genes conferring resistance to macrolide-lincosamide-streptogramin, β -lactams, and multidrug resistance²². At SIDI TOUAL and ANZA beaches, we observed aminoglycoside-related genes at 2,237 and 12,490 copies, respectively, and macrolide-related genes at 860 and 489 copies.

Among the most abundant ARGs are *aadA* (aminoglycoside adenylyltransferase), *ere* (macrolide esterases), *erm* (erythromycin resistance methylase), *Lnu* (lincosamide nucleotidyltransferase), *mph* (macrolide phosphotransferase), and *tet* (tetracycline-resistant ribosomal protection protein) (Fig. 4A). In February, prominent resistance genes at influent stations comprised *erm(B)* with 346,472 copies at ANZA WWTP, *mph(E)* at the same station with 510,895 copies, *strB/aph6* at TIZNIT with 154,573 copies, and *aadA13* also at the same station. In July, the prevailing genes remained *mph(E)* at DRARGA with 162,811 copies, *aadS* at the same station with 150,470 copies, *lnu(F)/lin(F)* at AOURIR with 95,659 copies, and *strB/aph6* at DRARGA with 81,777 copies. In the case of effluents, during the February period, the most abundant resistance genes consisted of *mph(E)* with 282,551 copies at ANZA, *strB/aph6* with 72,840 copies at AIT BAHA, and *lnu(F)/lin(F)* and *lnu(G)/lin(G)* with 80,774 and 70,823 copies at TIZNIT and AIT BAHA, respectively. Come July, the prevailing genes included *lnu(G)/lin(G)* with 99,197 copies at L'MZAR, *erm(F)* with 84,355 copies at TIZNIT, and *mph(E)* with

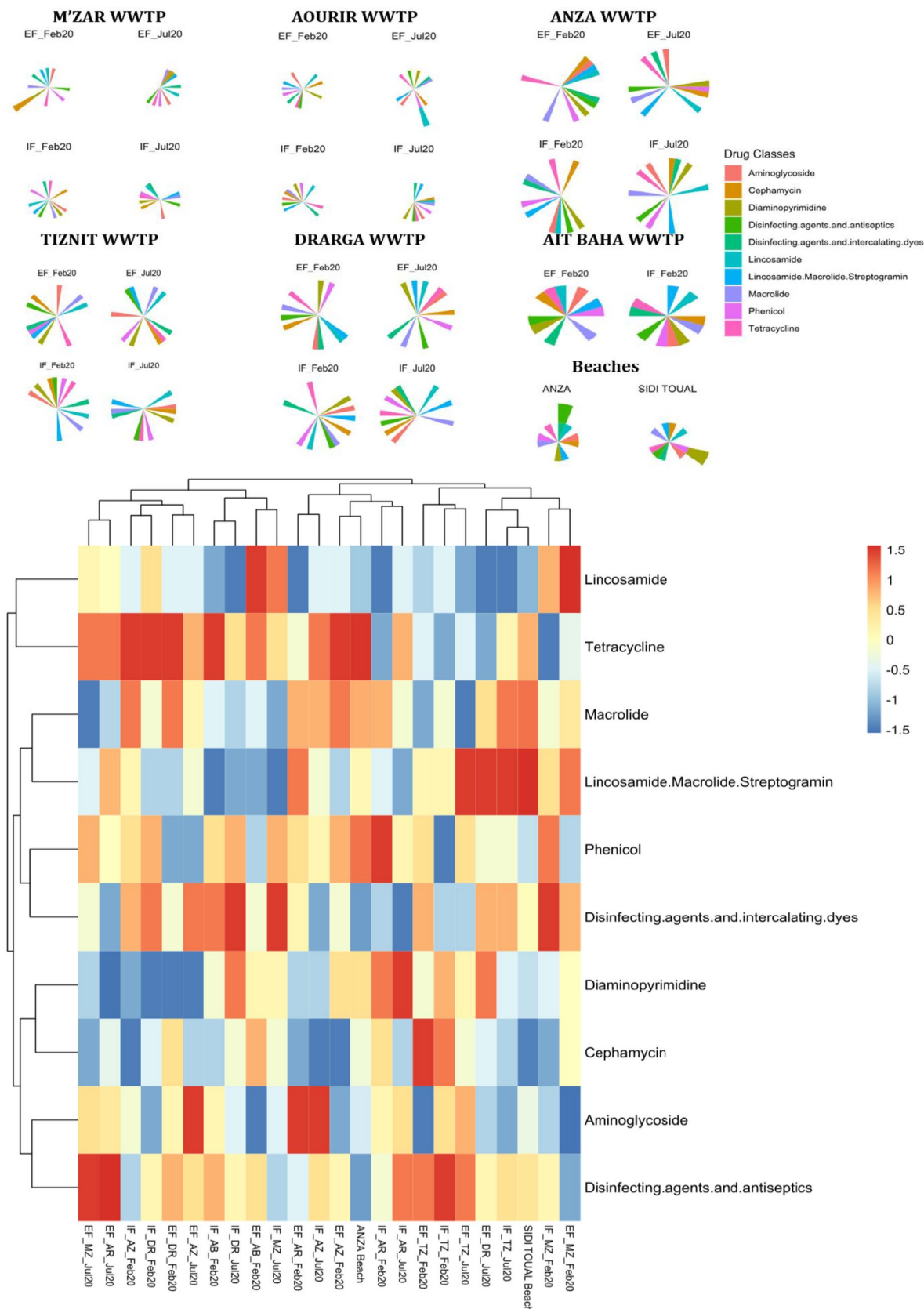


Fig. 3. The 10 most abundant drug classes.

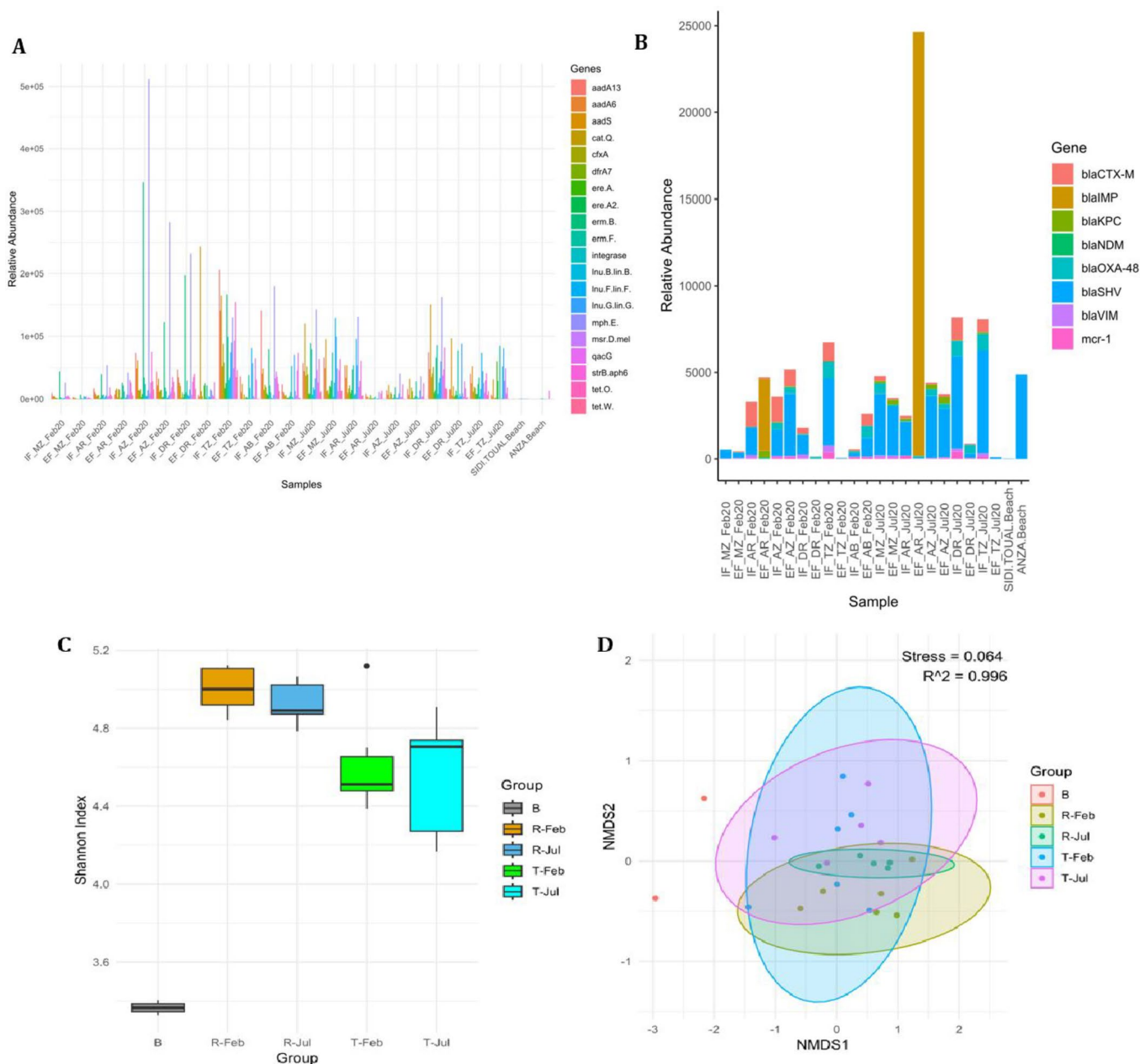


Fig. 4. **A.** Plot abundance of the 20 most abundant ARG, **B.** Plot abundance of other resistance genes, **C.** Alpha diversity using Shannon index, **D.** Beta diversity using Bray-Curtis method.

48,629 copies at L'MZAR. In the SIDI TOUAL and ANZA beaches, the most abundant genes are integrase with 1,032 copies (SIDI TOUAL), *strB/aph6* with 12,050 copies (ANZA), *mph(E)* with 585 copies (SIDI TOUAL), and *erm(F)* with 831 copies (ANZA).

The most predominant resistance genes in this investigation include *mph(E)*, *erm(B)*, *aadA*, and *strB/aph6*. The *mph(E)* resistance gene is typically associated with the encoding of a macrolide phosphotransferase enzyme, conferring resistance to macrolide antibiotics like erythromycin. Notably, this gene displayed a significant presence in the effluent originating from wastewater treatment plants near the Tama River and Tokyo²³. Its prevalence was universal, as it was detected in every sample during genomic analysis of sewage collected from 101 countries, providing a global perspective on antimicrobial resistance²⁴. Conversely, the *erm(B)* resistance gene encodes a ribosomal methyltransferase enzyme, conferring resistance to macrolide, lincosamide, and streptogramin B antibiotics. Notably, this gene stood out as the primary antimicrobial resistance gene identified in the influent of wastewater treatment plants²⁵. Furthermore, the *erm(B)* gene was quantified at significantly higher levels in the effluents of various municipal wastewater treatment plants across Germany, as highlighted in Norman's study²⁶.

Additionally, the *strB/aph6* gene, which often encodes a bifunctional enzyme in bacteria, possessing two distinct activities—streptomycin 3'-adenylyltransferase (*strB*) and kanamycin 6'-adenylyltransferase (*aph6*) was identified in an urban wastewater treatment plant^{27,28}.

For additional antimicrobial resistance genes quantified using qPCR, the copy count exhibited a marked increase in July 2020 as opposed to February. Among the prevalent genes detected in the majority of samples, the top four in terms of abundance and commonality are *blaSHV*, *blaCTX*, *blaOXA-48*, and *blaVIM* (Fig. 4B). The *blaSHV* and *blaCTX* genes stood out as the most prevalent, with *blaSHV* being the predominant one. These findings were consistently observed in numerous wastewater treatment plants worldwide²⁹. The number of copies fluctuates between 5 and 6000 copies. Regarding the *blaIMP* gene, its detection is primarily concentrated at the AOURIR station outlet during both time periods, exhibiting a copy number of 24,474 in the month of July. Ana's study shed light on its prevalence, notably showing that its relative abundance increased significantly with the size of the wastewater treatment plant, emphasizing its environmental impact³⁰. The *blaKPC* gene primarily manifests during the July timeframe, observable at the entrances and exits of M'ZAR and ANZA, as well as in the influents of AOURIR and TIZNIT. Concerning *mcr-1*, its predominant presence is observed primarily at the ANZA exit in February, registering 128 copies. Additionally, it is detected at the entrance of TIZNIT during both periods, with counts of 389 and 146 copies respectively. Notably, *mcr-1* is also identified at the AIT BAH station, with 118 copies detected at both the entrance and exit points, as well as at the DRARGA entrance in July, exhibiting a count of 441 copies.

The detection of *mcr-1* positive bacteria in influent wastewater samples from WWTPs in Germany, and in some cases, in effluent samples after conventional treatment, highlights the potential for these resistant bacteria to enter the aquatic environment²⁶.

The analysis of alpha diversity using the Shannon index revealed interesting insights into the richness and evenness of ARGs (Fig. 4C). The calculated Shannon index values showed that ARG richness decreases after the treatment of wastewater, with beach samples presenting the lowest index. In assessing beta diversity, we employed the Bray-Curtis dissimilarity metric (Fig. 4D). The findings indicated that the wastewater samples (both raw and treated) exhibited lower dissimilarity to each other compared to the beach samples.

When comparing all five groups (Raw wastewater of February: R-Feb, Raw wastewater of July: R-Jul, Treated wastewater of February: T-Feb, Treated wastewater of July: T-Jul, Samples of beaches: B) using PERMANOVA, a significant difference was observed with an F-statistic of 2.3779 and a p-value of 0.003 (**), indicating meaningful distinctions among the groups. However, when specifically examining the four groups (R-Feb, R-Jul, T-Feb, and T-Jul) through PERMANOVA, the analysis yielded an R-statistic of 1.3652 and p-value (> 0.05), suggesting no significant differences within these groups. Additionally, ANOSIM testing produced an R-statistic of 0.04707 and p-value (> 0.05), implying insignificant differences, potentially attributed to random variation.

Microbiome analysis

The 16 S primer pool enables the characterization of the microbiome with a resolution that extends to the species level. The Fig. 5A illustrates the 20 species with the highest levels of abundance in the collected samples (Fig. 5A). Predominantly featured among these species are *Acetobacterium sp*, *Achromobacter sp*, *Acidovorax sp*, and *Acinetobacter sp*. The ANZA station exhibits the highest diversity in terms of both entry and exit during the month of February. In the inflow of DRARGA and TIZNIT stations in July, *Acetobacterium sp* is primarily observed at relative abundance of 0.05%. *Achromobacter sp* is predominantly detected at the outlet of the TIZNIT station in both July and February, with relative abundance of 0.08% and 0.04%, respectively. *Acidovorax sp* is predominantly observed in the effluents of AOURIR during both February and July, accounting for 0.2% of the microbial community in February and 0.11% in July. It is also detected in the effluent of TIZNIT in July, constituting 0.1% of the microbial community. *Acinetobacter sp* is primarily detected in the inflow of ANZA during February, making up 1.18% of the microbial community. It is also observed in the effluent of the same station during the same period, accounting for 0.8% of the microbial community.

A total of twenty-one species, which possess the capacity to spread resistance within the environment and potentially transmit it to humans, including the ESKAPE pathogens, have been identified. Out of these, 18 species have been observed (Fig. 5B). The species predominantly detected across all samples include *Streptococcus salivarius*, *Escherichia coli*, *Klebsiella pneumoniae*, and *Citrobacter freundii*. Notably, *Streptococcus salivarius* was observed at the ANZA station in February, accounting for 7% at the outlet and 5% at the inlet. *Escherichia coli* was primarily identified at the same station, comprising 13% at the inlet and 4% at the outlet. *E. coli*, has emerged as a major concern due to its increasing antibiotic resistance. This resistance not only hampers the effectiveness of conventional therapies but also elevates the risk of prolonged illnesses, healthcare-associated infections, and broader community dissemination³¹. *Klebsiella pneumoniae* was primarily detected at the M'ZAR station inlet in July, the DRARGA inlet in February, and the ANZA outlet during the same period, with a relative abundance of 2%. Despite this, the pathogenic bacteria were eliminated after treatment at both M'ZAR in July and Drarga in February. In contrast, at the ANZA site, treatment was less effective, as the bacteria persisted in the February effluent and even showed an increase post-treatment during this period. *Klebsiella pneumoniae* is among the most prevalent gram-negative strains commonly found in hospital wastewater, heightening concerns regarding its role in the development and spread of antibiotic resistance genes³². Moreover, *Klebsiella pneumoniae* is considered a key trafficker of drug resistance genes from environmental to clinically important bacteria³³. The persistence of such bacteria, especially in effluents, underscores the challenges in controlling antibiotic-resistant pathogens and the need for more effective wastewater treatment processes.

Citrobacter freundii was detected at the M'ZAR inlet in July, constituting 2.6%, but was eliminated after treatment. At the ANZA outlet in February, it comprised 1.4%, yet it was not present during the July period. In February, the ANZA inlet and outlet exhibited the highest prevalence of these species, whereas in July, the

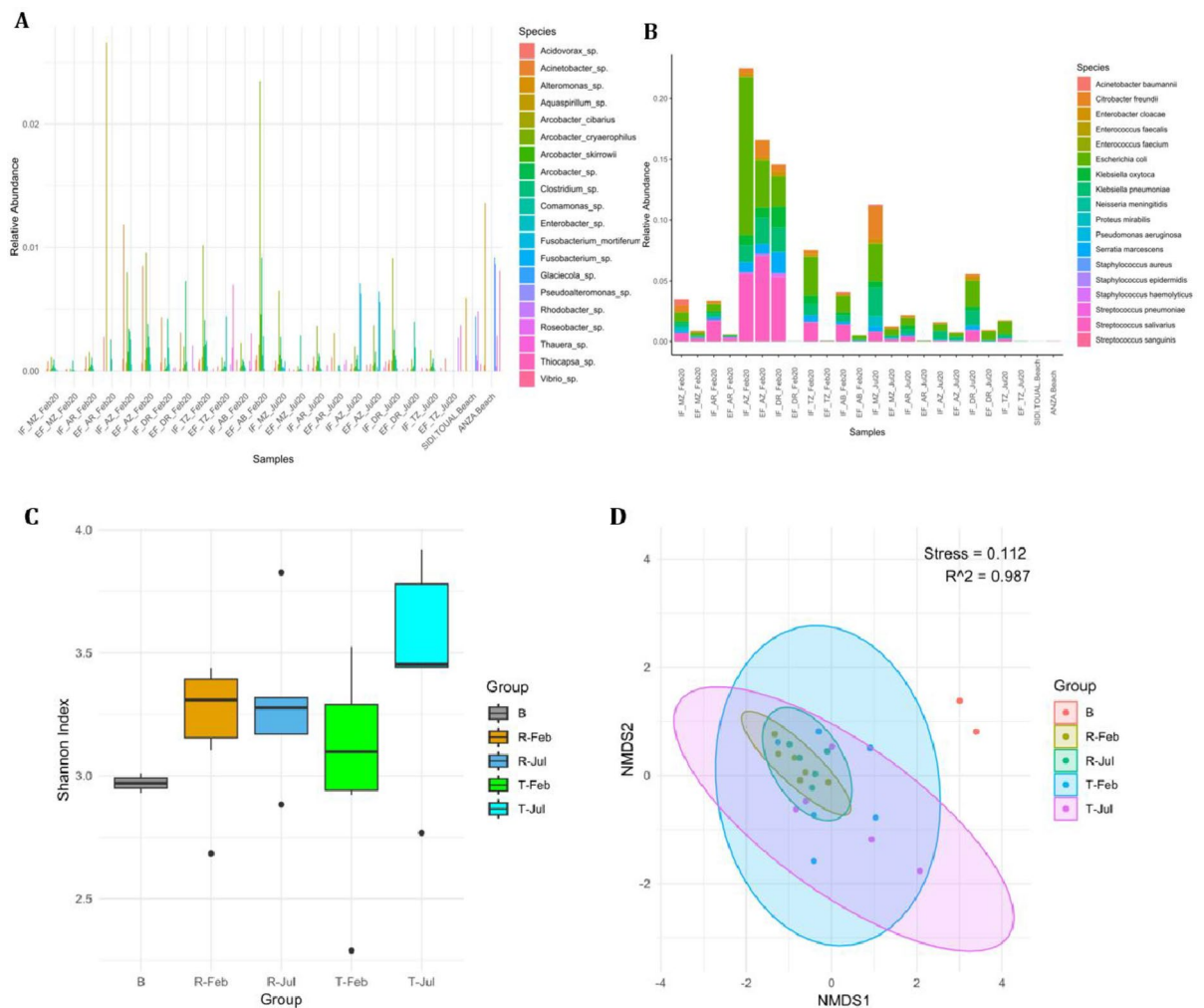


Fig. 5. **A.** Plot abundance of the 20 most abundant species, **B.** Plot abundance of other species, **C.** Alpha diversity using Shannon index, **D.** Beta diversity using Bray-Curtis method.

M'ZAR and DRARGA inlets displayed the highest occurrence. Regarding the other bacteria mentioned in Fig. 5B, they are nearly absent.

The Shannon index values indicated that treated wastewater in July exhibited higher diversity compared to both raw wastewater and treated wastewater in February (Fig. 5C). Using the Bray-Curtis method, the analysis revealed less dissimilarity between raw and treated wastewater samples, while beach samples demonstrated marked dissimilarity, highlighting the distinct microbial communities present in the natural environment (Fig. 5D). A PERMANOVA test comparing all five groups revealed a significant difference, with an F-statistic of 2.411 and a p-value of 0.001 (*), indicating meaningful distinctions across these groups. Narrowing the focus to four groups (R-Feb, R-Jul, T-Feb, and T-Jul), the PERMANOVA analysis showed a significant variation with an R-statistic of 1.7331 and a p-value of 0.01, underscoring the differences between raw and treated wastewater across months. Further statistical analysis using ANOSIM yielded an R-statistic of 0.1534 and a p-value of 0.017, confirming significant differences between the groups. The observed variations are linked to the nature of the influent, whether it is domestic (urban or rural) or industrial, as well as the wastewater treatment process employed. Additionally, the presence of pathogenic bacteria from the ESKAPE group in the effluent, such as at the ANZA WWTP and ANZA Beach, raises significant concerns. These findings indicate that while treatment reduces the overall bacterial load, the microbial community composition changes considerably with seasonal fluctuations and the specific treatment processes used, particularly when compared to natural environments like beach samples.

Efficiency of WWTPs

The examined wastewater treatment facilities employ various treatment methods. The M'ZAR plant utilizes anaerobic decantation, whereas the AOURIR plant employs activated sludge. The ANZA plant employs lamellar decantation, while both the TIZNIT and AIT BAHIA plants resort to natural lagooning. The final DRAGARA plant utilizes infiltration and percolation over sand. The efficacy of the employed treatments was assessed at all stations during both time periods. The formula employed for evaluation was ((influent_concentration -

$\text{effluent_concentration} / \text{influent_concentration} \times 100$. This assessment pertains to the purification efficiency of four constituents: ARGs, ORGs (other resistance genes), the microbiome, and the species accountable for the propagation of resistance (Fig. 6).

It is important to highlight that there is a variation in treatment efficacy when comparing the two periods. The optimal approach for ARG elimination was observed at the AOURIR facility in July, achieving an 86% reduction, closely trailed by TIZNIT in February at 81%. Conversely, AOURIR in February and ANZA in July exhibited subpar efficacy. In terms of ORG elimination, the TIZNIT treatment exhibited exceptional performance with a consistent 98% reduction in both periods. Notably, the AOURIR station displayed reduced effectiveness due to the prevalence of the *blaIMP* gene in its effluents. Furthermore, the AIT BAHA station also registered an elevation in *mcr-1*, *blaOXA-48*, *blaSHV*, and *blaVIM* genes within the effluents.

Regarding the eradication of resistance-spreading species, TIZNIT demonstrated a remarkable 99% reduction in both periods, while DRARGA achieved the same percentage in February. Additionally, the AOURIR station exhibited a 96% reduction in July. Conversely, the ANZA station recorded the lowest elimination percentages, with 26% in February and 53% in July. Regarding the final element of the microbiome, M'ZAR achieved the highest elimination percentage in February, reaching 62%. However, certain stations exhibited limited effectiveness in microbiome elimination, leading to an effluent increase. This was particularly noticeable at AOURIR and AIT BAHA in February, as well as TIZNIT and DRARGA in July. In summary, it can be asserted that the efficacy of a treatment is contingent upon numerous variables, encompassing influent quality, composition, physicochemical parameters, seasonal variations, and potentially undisclosed factors.

Discussion

This study provides a spatio-temporal analysis of the resistome and microbiome in wastewater treatment plants in Agadir, Morocco, using a metagenomic approach. Six facilities with varying treatment processes were sampled during February and July. The results show that the predominant resistance genes are related to aminoglycosides, LMS, macrolides, and lincosamides. Twenty-one species with the potential to disseminate resistance throughout the environment and potentially transfer it to humans have been identified. Out of these, 18 species have been observed. The species predominantly detected across all samples include *Streptococcus salivarius*, *Escherichia coli*, *Klebsiella pneumoniae*, and *Citrobacter freundii*.

The 16 S primer pool enables the characterization of the microbiome with a resolution that extends to the species level. Predominantly featured among species are *Acetobacterium sp.*, *Achromobacter sp.*, *Acidovorax sp.*, and *Acinetobacter sp.*

The effectiveness of wastewater treatment depends on various factors, including the quality and composition of the influent, physicochemical parameters, seasonal variations, and potentially other undisclosed variables. In our study, the most efficient WWTPs were Aourir, which employs an activated sludge process with aerobic treatment, and Tiznit with a lagooning system. Multiple studies have investigated different treatment strategies commonly used in WWTPs for their efficiency in removing antibiotic resistance genes (ARGs) and antibiotic-resistant bacteria.

For instance, aerobic and anaerobic-aerobic systems generally demonstrate better ARG removal compared to anaerobic-only systems³⁴. The removal efficiency of these systems varies between 33% and 85%, though some studies noted an increase in the relative abundance of certain ARGs^{35–37}. To achieve higher removal rates, up to 99%, membrane bioreactors³⁸ and advanced disinfection methods like chlorination, ozonation, and UV irradiation have proven effective³⁹, though some of these methods can be costly. Constructed wetlands and hybrid systems have also shown promising results in removing ARGs and resistant bacteria⁴⁰.

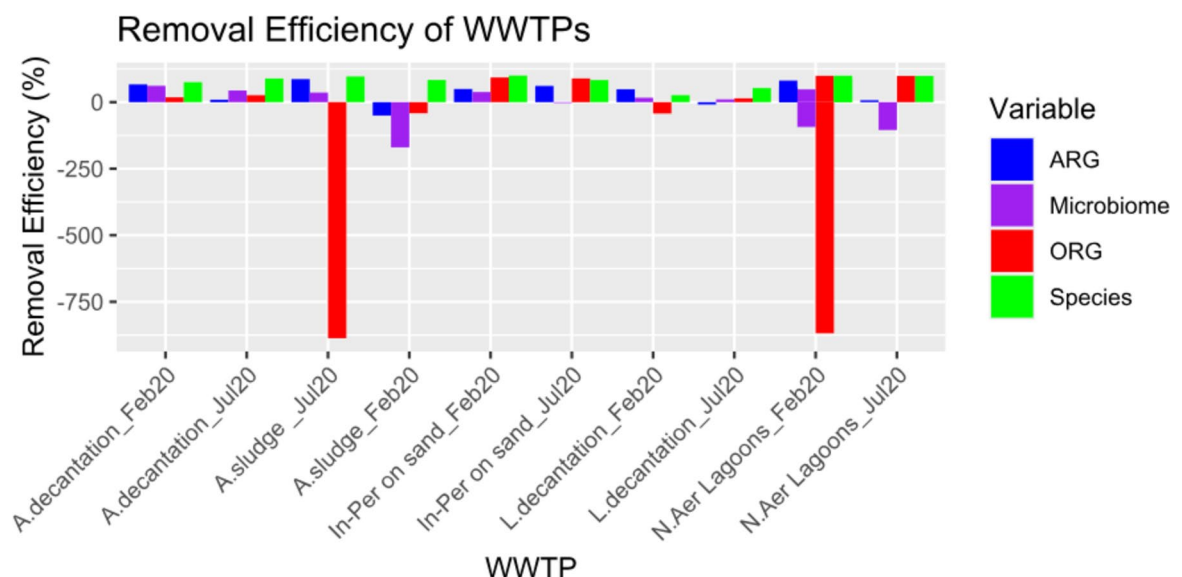


Fig. 6. Removal Efficiency of WWTPs.

For comprehensive reviews on the efficacy of these treatment methods, please refer to Uluseker et al.⁴¹ and Huang et al.⁴²

The emergence of antibiotic resistance poses a significant global public health challenge, as bacterial infections are a leading cause of mortality. Currently, there were around 4.95 million (3.62–6.57) deaths associated with bacterial resistance in 2019, of which 1.27 million (95% UI 0.911–1.71) were attributed to resistant bacteria⁴³. This resistance can lead to therapeutic impasse. Several healthcare establishments have recorded an increase in nosocomial infections, mainly due to bacteria producing extended-spectrum B-lactamases (ESBL)⁴⁴. The epidemiology of this phenomenon shows a worrying growth, especially with the excessive use of antibiotics in various fields, such as the treatment of infections in humans or animal husbandry. This phenomenon is exacerbated by the existence of several means of dissemination, notably conjugative plasmids, transposable elements and integrons, which enable the genes conferring resistance to be moved between different bacteria⁴⁵. In addition, repeated administration of antibiotics and non-compliance with dosages encourage selection pressure and cause the acquisition and dissemination of drug-resistant strains. The World Health Organization (WHO) considers infections caused by these bacteria to be a priority for treatment, due to their potential impact on public health. It can even affect food security and influence the development of countries. This phenomenon is currently affecting several countries to varying degrees, especially in the wake of the COVID-19 pandemic. Epidemiological monitoring of this resistance is an important means of limiting the spread of resistant bacteria and implementing mitigation strategies. Such biomonitoring requires global action to establish a standardized method for comparing data from different countries⁴⁶. This approach enhances the analysis and communication of results, enabling informed decision-making. Biomonitoring underscores the importance of proper antibiotic use in both human and animal health, adherence to hygiene practices, and the development of strategies to maintain antibiotic efficacy.

Canonical Correspondence Analysis (CCA) was used to explore how various physicochemical parameters relate to the presence of antibiotic resistance genes (ARGs). Factors like pH, Temperature, Dissolved O₂, Conductivity, Salinity, Turbidity, COD, Calcium, TH, Magnesium, TAC, Chlorides, and AnSb were studied as independent variables, while the abundance of ARG was the dependent variable. Results indicated that these physicochemical parameters collectively explain a significant portion of the variation in ARG composition (CCA1: $p=0.007$), suggesting a meaningful link between the overall physicochemical environment and antibiotic resistance gene distribution. While CCA1 showed significance, subsequent axes (CCA2 to CCA13) didn't display statistically significant connections with ARG composition. These findings highlight the role of specific physicochemical parameters in shaping ARG profiles and emphasize the need for further research into the individual parameters that drive this relationship. It is important to note that although CCA identifies associations, more research is needed to understand the underlying mechanisms and potential causal factors behind these observed relationships. For the relationship between other resistance genes and physicochemical parameters using Canonical Correspondence Analysis, the research revealed that these parameters accounted for a substantial portion (about 85.6%) of the observed variability in ORG prevalence. Notably, CCA1 showed significant correlation ($p<0.05$) with the physicochemical factors, underlining their role in influencing ORG composition. Eigenvalue analysis demonstrated that CCA1 was a key contributor, explaining approximately 39.85% of the total variance. These findings shed light on the intricate connections between ORGs and environmental conditions. The association between antibiotic resistance genes (ARGs) and mobile genetic elements (MGEs), particularly integrase and the species accountable for resistance dissemination, demonstrated robust correlations (Spearman correlation of 0.82 and 0.87, respectively) (Fig. 7).

In contrast, the correlation with the microbiome exhibited a lesser significance at a value of 0.46. Similarly, a robust correlation of 0.81 was observed between MGEs and the species responsible for disseminating resistance. The figure below provides a detailed representation of the co-occurrence between the resistome and microbiome (Fig. 8).

Conclusion

Water scarcity is an urgent global challenge. The reuse of wastewater poses numerous hazards despite offering a potential solution. The analysis of the resistome and microbiome within wastewater treatment plants revealed the emergence of numerous genes, notably *mph(E)*, *erm(B)*, *aadS*, *strB/aph6*, *blaSHV*, and *blaCTX*. Pathogens identified include *Escherichia coli*, *Streptococcus salivarius*, *Klebsiella pneumoniae*, *Arcobacter cryaerophilus*, *Clostridium* species, and *Acinetobacter* species. High levels of mobile genetic elements were also observed. The coexistence of MGE and antimicrobial resistance genes underscores their pivotal role in facilitating the acquisition and dissemination of ARGs within microbial communities through horizontal gene transfer. Several purification methods have demonstrated effectiveness in removing a significant portion of these genes and their associated pathogens. However, additional treatment is necessary to completely eliminate these contaminants, ensuring that the water can be safely utilized without posing health risks.

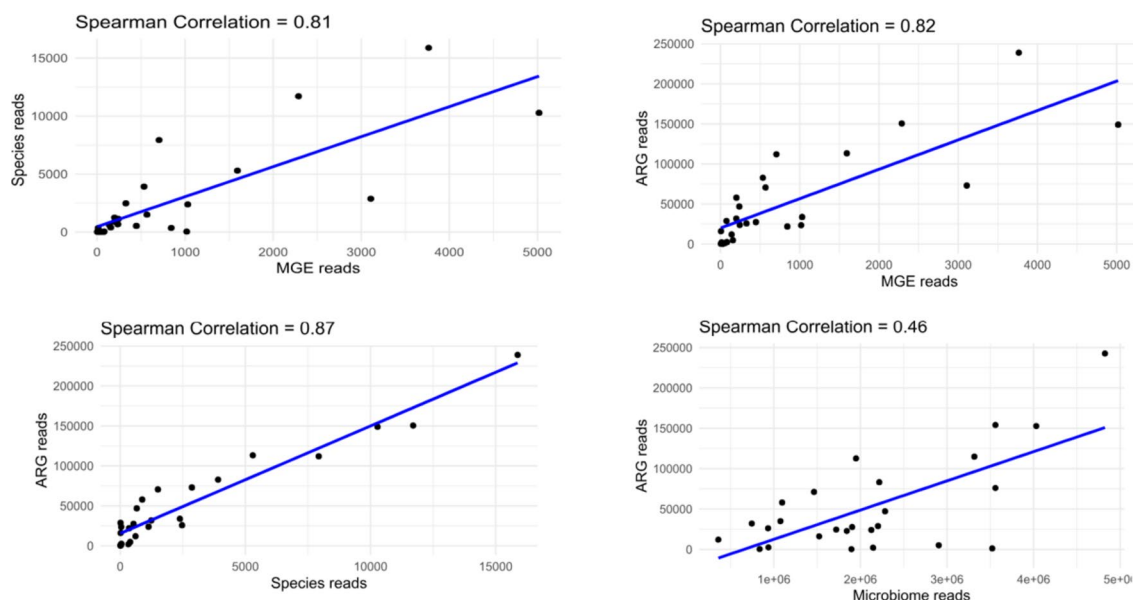


Fig. 7. Spearman correlation analysis among Mobile Genetic Elements (MGE), Antibiotic Resistance Genes (ARG), microbiome and species.

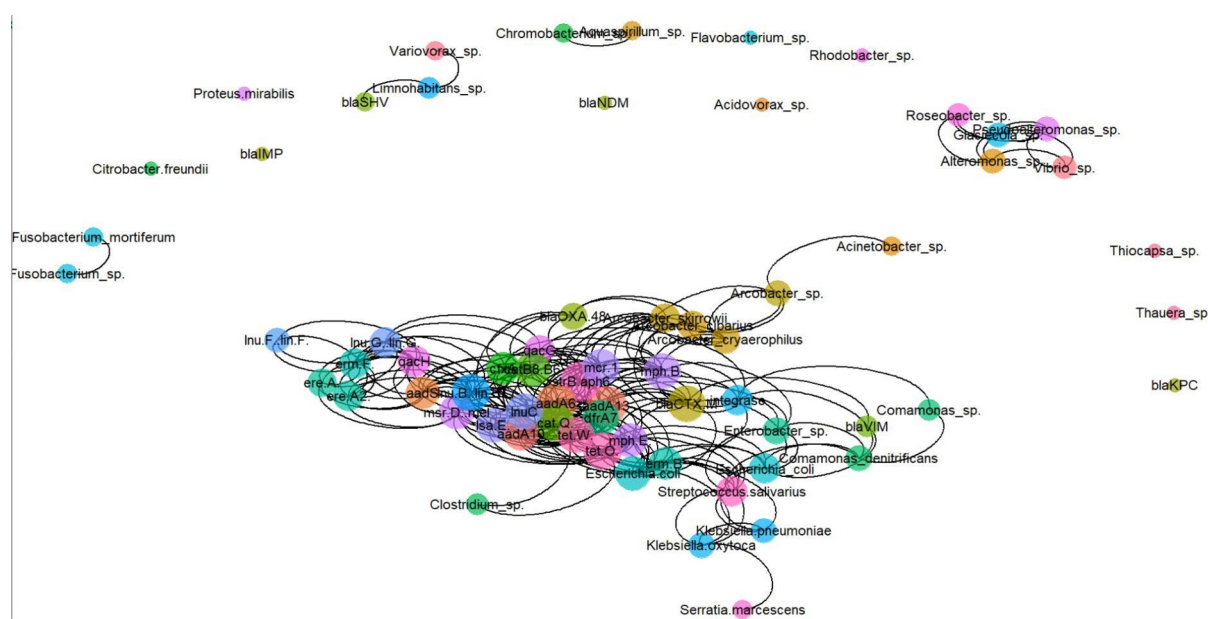


Fig. 8. Resistome and Microbiome co-occurrence network analysis.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Maryem Wardi : Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. Zohra Lemkhente : Conceptualization, Methodology, Validation, Formal analysis, Writing - original draft, Writing - review & editing. Aicha Ait Alla : Conceptualization, Methodology, Resources. Nouredine Slimani : Conceptualization, Resources. M'hamed Abali : Conceptualization, Resources. Youssef Idaghdour : Conceptualization, Methodology, Formal analysis, review & editing. Ahmed Belmouden : Supervision, Conceptualization, Resources, Writing - original draft, Writing - review & editing.

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

Not applicable.

Additional information

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